MILITARY SPECIFICATION

SWITCH, STEPPING, DIRECT CURRENT, AIRCRAFT DISPENSER, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for the manufacture and inspection of one type of stepping switch.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-P-455	-	Plates and Foils, Photographic (Photosensitive Anodized Aluminum).
QQ-S-571	-	Solder, Tin Alloy, Lead-Tin Alloy and Lead Alloy.
TT-L-20	-	Lacquer, Camouflage.
TT-P-1757	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity.
ZZ-R-765	-	Rubber, Silicone.

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MIL-I-10	-	Insulating Materials, Electrical, Ceramic, Class L.
MIL-M-14	-	Molding Plastics and Molded Plastic Parts, Thermosetting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-997	-	Plastic Material, Laminated, Thermosetting, Electrical Insulation, Sheets, Glass Cloth, Silicone Resin.
MIL-E-5400	-	Electronic Equipment, Aerospace, General Specification for.
MIL-A-8625	-	Anodic Coatings, For Aluminum and Aluminum Alloys.
MIL-F-14256	-	Flux, Soldering, Liquid (Rosin Base).
MIL-P-15024	-	Plate, Tags and Bands for Identification of Equipment.
MIL-P-15024/2	-	Plate, Identification, Unit or Plug-in Assembly.
MIL-P-15037	-	Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin.
MIL-P-15047	-	Plastic Material, Laminated Thermosetting Sheets, Nylon Fabric Base, Phenolic-Resin.
MIL-N-18307	-	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment, Including Ground Support Equipment.
MIL-W-22248	-	Weldment, Aluminum and Aluminum Alloy.
MIL-C-26482	-	Connector, Electrical, (Circular, Miniature, Quick Disconnect, Environment Resisting) Receptacles and Plugs, General Specification for.
MIL-I-43553	-	Ink, Marking, Epoxy Base.
MIL-S-45743	-	Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment.
MIL-I-46058	-	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies).
MIL-S-85377/1	-	Switch, Stepping, One Level, Eight Position.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards for Federal Services.

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DOD-STD-100	-	Engineering Drawing Practices.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-143	-	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-704	-	Aircraft Electric Power Characteristics.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-831	-	Test Reports, Preparation of.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-965	-	Parts Control Program.
MIL-STD-1168	-	Lot Numbering of Ammunition.

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1385 - Preclusion of Ordnance Hazards in Electromagnetic Fields; General Requirements for.

HANDBOOKS

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MIL-HDBK-235-2 - Electromagnetic (Radiation) Environment from Friendly or Own Force Emitters.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND

AR-10 - Maintainability of Avionics Equipment and System, General Requirements for.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. The switch furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.3).

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Interchangeability. All parts having the same part number shall be directly and completely interchangeable with each other with respect to installation and performance. The switch shall meet the interchangeability requirements of MIL-E-5400, and the part numbers shall be governed by DOD-STD-100.

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3.5 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to MIL-E-5400, except that conformal coatings shall be in accordance with MIL-I-46058.

c. Nonrepairable subassemblies shall be used in accordance with AR-10 and as outlined in MIL-E-5400.

3.5.1 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials (including transistors and diodes) other than microelectronic devices shall be obtained as outlined in MIL-E-5400. Microelectronic devices shall be approved as outlined in MIL-STD-965.

3.5.2 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall meet the requirements of MIL-STD-965. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.6 Metals. All metal parts (other than current-carrying parts) exposed to environmental conditions shall be of corrosion-resistant material, or shall be suitably plated to resist corrosion during service life.

3.6.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals, as defined in MIL-STD-889, shall not be used in contact with each other.

3.6.2 Fungus proof materials. All materials shall be non-nutrients for fungi.

3.7 Insulators.

3.7.1 Ceramic insulators. Ceramic insulation shall be Grade L411 or higher, in accordance with MIL-I-10.

3.7.2 Plastic insulators. Laminated-plastic sheets shall conform to MIL-P-997, MIL-P-15037 or MIL-P-15047.

3.7.3 Molded plastic. Molded plastic material shall conform to MIL-M-14, except that cotton, wood, or cellulose-filled molding materials shall not be used.

3.7.4 Silicone rubber. All silicone rubber parts shall be in accordance with ZZ-R-765.

3.8 Spring embrittlement. Springs used in the fabrication of the switch shall not break when extended or distorted to the point of minimum permanent distortion.

3.9 Threaded parts. All threaded parts shall be in accordance with FED-STD-H28.

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3.9.1 Engagement of threaded parts. All threaded parts used for mounting means shall engage for a minimum of three full threads.

3.10 Soldering. All soldering must be done in accordance with MIL-S-45743.

3.10.1 Soldering and soldering flux. When soldering is employed, all flux shall be removed after soldering. Electrical connections shall be mechanically secure and electrically continuous before and after soldering. Solder, when used, shall be in accordance with QQ-S-571 and soldering flux shall be in accordance with MIL-F-14256.

3.11 Welding. Welding procedures shall be in accordance with MIL-W-22248, Class III.

3.12 Housing. The switch shall have a water tight metal case. The case shall not be used for a current-carrying conductor but it may be used as part of the magnetic circuit or shield. The finish around the mounting holes shall provide positive electrical contact to ground when the switch is mounted.

3.13 Finish. The switch housing shall be finished as listed in the specification sheet.

3.13.1 Painting limitations. Neither primer nor paint shall be applied to threads, bearings or working surfaces.

3.14 Total weight. The total weight of the switch, excluding cables, shall be kept to a minimum, consistent with good design and shall not exceed that listed in the specification sheet.

3.15 Cabling and connections. The switch shall use internal cabling, printed wiring, and connections in accordance with MIL-E-5400.

3.15.1 Connectors. The switch shall mate with the applicable mating connector specified in the specification sheet, and shall meet the requirements of MIL-C-26482.

3.15.2 Interconnection cabling. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the switch. External cables shall be unshielded.

3.16 Terminals. Terminals shall not be used for external connections.

3.17 Switch position indicator label. A switch position indicator label, as specified in the specification sheet, shall be used.

3.18 Design and construction. The design and construction of the switch and parts thereof shall be in accordance with this specification and the applicable specification sheet. The switch shall be capable of being assembled into an airborne dispenser and controlling its firing sequence under all flight and environmental conditions.

3.19 Operational stability. The switch shall operate to specified performance, continuously or intermittently, for a period equal to the operating life (see 3.26) without the necessity for readjustment.

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3.20 Operational characteristics. The switch, located within an aircraft weapon dispenser shall distribute the incoming firing signal in sequence to each impulse ejection cartridge in the dispenser breech assembly. The voltage may be interrupted as a result of any type of switch contact bounce for a maximum of 5.0 milliseconds (MS) upon application of power without interfering with the proper operation of the switch.

3.20.1 Safe circuit. The switch shall have a safe and an arm position. The safe position shall internally ground all output circuits. The internal grounding shall be accomplished by means of the ground circuit, 3.20.5. The pulse following the last firing pulse shall advance the switch into the safe position. The switch shall not be capable of being electrically advanced out of the safe position. The switch shall be advanced manually to arm.

3.20.2 Arm circuit. The arm circuit shall provide a ground circuit to all outputs, but shall also allow for electrical sequencing. The switch shall be manually switched from the safe position to the arm position to provide positive arming of the unit.

3.20.3 Sequencing. The switch shall sequence through the output positions as called out in the specification sheet. The output positions may or may not be loaded. The switch shall continue through the sequence pattern each time power is applied and then removed. The time between application of power and output pulse shall not exceed 30 MS.

3.20.4 Bypass capability. The switch shall, in the event of a short circuit to ground or an open circuit on output pin(s), attempt to fire and in the case of an open circuit remain in place. The next application of power will sequence the switch to the next position. The switch bypass capability shall withstand short circuit current without damage.

3.20.5 Ground circuit. A ground circuit shall be provided through the switch to ensure safety during loading and operation. The ground circuit shall ground all output pins when the switch is in the safe or arm position. As the switch is electrically advancing through its sequence, the ground circuit shall individually ground each output pin, except those being energized. The ground circuit shall ground the output with a resistance of not more than 0.2 OHMS.

3.21 Electrical characteristics (see 4.4.1.2).

3.21.1 Electrical wiring. There shall be no failure of electrical requirements due to handling or environmental conditions.

3.21.2 Warm-up-time. The switch shall not require a warm-up-time and must be ready for full operation upon application of power.

3.21.3 Ground. The ground connector pin is the negative return for the switch and must be capable of carrying 10 amperes current for one-minute duration. The external parts of the switch shall be at ground potential at all operating voltages, regardless of selector switch position. Ground circuit resistance (switch in safe or arm position) shall be 0.2 OHM maximum between any firing pin and ground.

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3.21.4 Unused pins. The unused pins must have at least one megohm resistance to any other input or output and must be available for future use.

3.21.5 Input parameters. The switch shall operate in accordance with all requirements when input power of 24.5 to 31.5 VDC is supplied through a total series input resistance, which can range in value from 2.5 OHMS minimum to 7.2 OHMS maximum, and a total line inductance up to 200 microhenrys maximum. It shall also operate in accordance with all requirements when input power of 22.0 to 24.0 VDC is supplied through a 5.0 ± 0.1 OHM current-limiting resistor of suitable power rating. Other input parameters are given in the specification sheet.

3.21.5.1 Input voltage rise time. The input voltage rise time shall be 10.0 MS or less (straight line from +10 percent to +90 percent of input voltage).

3.21.5.2 Dropout voltage. The dropout voltage, the voltage that the switch considers an off voltage, shall be +0.5 VDC minimum at any input over the temperature range -55°C to +71°C.

3.21.6 Timing.

3.21.6.1 Initial delay time. The initial delay time is that interval between application of rated voltage and the occurrence of a firing output. The initial delay time shall include contact bounce appearing on any firing output line. No output shall occur at any previously fired station during the initial delay time.

3.21.6.2 Externally-pulsed stepping speeds. The switch shall be capable of following an essentially square wave pulse at 22 through 31.5 VDC having a minimum on-time and a minimum off-time as specified in the specification sheet.

3.21.6.3 Dwell time. The dwell time is the time the firing output current remains on a loaded output position assuming continuous application of input power (see 4.9.2.1). Dwell time is specified in the specification sheet.

3.21.6.4 Contact bounce. When the switch is tested as specified in 4.9.2.4, contact bounce appearing on any firing output line shall be as specified in the specification sheet.

3.22 Environmental. The switch shall meet all of the requirements of 3.20 and 3.21 before, during, or after exposure to the following environmental service conditions or reasonable combination of these conditions:

a. Humidity (see 4.7.1).

b. Salt fog (corrosion) (see 4.7.2).

c. Sand and dust (see 4.7.3).

d. Barometric pressure (reduced) (see 4.7.4).

e. Thermal shock (see 4.7.5).

f. Shock (see 4.7.6).

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g. Acceleration (see 4.7.7).

h. Vibration (see 4.7.8).

3.23 Electromagnetic compatability. When installed in a dispenser, circuit protection from electromagnetic interference (susceptibility) shall be provided in the switch in accordance with MIL-STD-461 (Class ID equipment). The switch, installed in a dispenser, shall be able to withstand subjection to the electromagnetic (radiated) environments found in MIL-HDBK-235-2, Table VA, and MIL-STD-1385.

3.24 Simulated firing. The switch shall meet all requirements of 3.20 and 3.21 when the firing load is simulated by use of a dummy load consisting of a resistor connected between each firing pin and ground.

3.25 Operating power. The switch shall exhibit the specified performance when energized from a 27.5 VDC power source having characteristics and limits as defined in MIL-STD-704. The transient input voltage at any input shall be as specified in MIL-STD-704 with the following limits:

a. Peak transient voltage

Peak transient voltage (volts)	Maximum transient pulse length (MS)	Minimum series input resistance (OHM)
+60	20	5.7
+50	45	5.0
+40	90	2.5

b. Spike peak: ± 200 volts for 5 microseconds (50 OHM source, energy 0.01 joule).

3.26 Operating life. The switch shall meet all the requirements of 3.20 and 3.21 after a minimum of 500 firing cycle operations and 500 manual arming operations when tested and accepted as outlined under the requirements of 4.8 and Tables I and III.

3.27 Identification marking. The switch shall contain the markings and identification plate specified in the specification sheet, or in accordance with MIL-STD-130, if not specified in the specification sheet. Any change in part number shall be governed by DOD-STD-100.

3.27.1 Nomenclature, nameplates, and identification marking. Nomenclature and serial number assignment, nameplate approval, and identification markings shall be in accordance with MIL-N-18307.

3.28 Workmanship. The workmanship displayed in fabrication and assembly of the switch shall be such as to assure, within design limitations, the ability of the switch to meet performance requirements under all applicable environmental conditions specified herein. Unauthorized repair, welding, heavy burrs, or parts

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assembled by introduction of high stresses not prescribed in the drawing are typical signs of inferior workmanship and shall be cause for rejection. The standards of workmanship exhibited in the approved first article sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship.

4. QUALITY ASSURANCE PROVISIONS (see 6.4)

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of all the inspections listed in Table I. First article inspection report format shall be in accordance with MIL-STD-831 (see 6.3).

4.3.1 First article samples. First article inspection shall be made on sample switches representative of the production units to be supplied under contract. The first article sample shall consist of twelve (12) switches, using the same production procedures and equipment which will be used in fulfilling the contract. Each switch shall be permanently labeled for ease of identification. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production in excess of 90 days, or whenever a change occurs in manufacturing, material used, drawing or specification such as to significantly affect production uniformity as determined by the Government acquiring activity. Failure of any switch to pass any of the first article inspections shall be cause for rejection.

4.3.2 Scope of inspections. The scope of first article inspection shall include all tests listed in Table I. The switch shall be operated at the beginning and conclusion of each service condition to determine whether a failure occurred during the service condition. Additional operation during testing shall be made when specified by Table I or by the appropriate test method. The outputs shall be monitored during operation to verify that proper output voltages are present only when the prescribed inputs are present. The quantity of switches required for each test is listed in Table I.

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4.3.3 First article approval. Approval of the first article sample shall be by the acquiring activity upon satisfactory completion of all tests and examinations.

4.3.4 Production switches. Switches supplied under the contract shall, in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved first article sample. Each switch shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of noncompliance with the above shall constitute cause for rejection.

4.4 Quality conformance inspection. The contractor shall furnish all samples (quantity as per 4.4.2.4) and shall be responsible for accomplishing the quality conformance inspection. All inspection and testing shall be under the supervision of the Government quality assurance representative/specialist. The contractor shall furnish test reports showing quantitative results for all quality conformance inspection tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Quality conformance inspection report format shall be in accordance with MIL-STD-831 (see 6.2.2). Quality conformance inspection shall consist of the following:

a. Individual inspections (see 4.4.1).

b. Lot inspections (see 4.4.2).

4.4.1 Individual inspections. Each switch submitted for acceptance shall be subjected to the individual inspections. These inspections shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. Individual inspections for the switches are specified in Table II. As a minimum, each switch shall have passed the following:

a. Physical inspection (see 4.4.1.1).

b. Electrical test (see 4.4.1.2).

c. Manufacturing run-in test (see 4.4.1.3).

Failure to pass any of these inspections shall be cause for rejection of the switch.

4.4.1.1 Physical inspection. Each switch shall be examined to verify that materials, design, construction, physical dimensions, marking and workmanship are as specified herein as well as to determine conformance to the requirements of this specification and applicable documents for which no specific tests are described (see 3.28). Each switch shall be examined to ensure that the packaging, packing and marking conforms to this specification (see Section 5).

4.4.1.2 Electrical tests. Each switch shall be operated using the tests of 4.9.7, 4.9.3 and 4.9.4 to check characteristics and record adequate data to assure satisfactory switch operation.

4.4.1.3 Manufacturing run-in tests. Each switch shall be operated under the conditions specified herein without failure. A failure is defined as any malfunction of the switch. No adjustments will be permitted during the normal use of the switch. The following environmental conditions apply to this test:

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Input voltage	As specified in the specification sheet
Temperature	Ambient room
Humidity	Ambient room
Vibration	None - Unless required by contract

The switch shall be operated for a minimum of one (1) cycle of each of the operational tests in 4.9.7 without failure. Failure of a switch shall be cause for rejection of the switch.

4.4.2 Lot (sampling) inspections. Lot (sampling) inspections shall be as specified in Table III. Failure to pass any of these inspections shall be cause for rejection of the entire lot (see 6.2.1).

4.4.2.1 Lot formation. The term, "inspection lot", as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements (see MIL-STD-109). The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings and same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability or storage, as approved by the Government acquisition activity, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall bear a unique lot number.

4.4.2.2 Inspections. The inspections listed in the following subparagraphs shall be performed on inspection lots (see 4.4.2.1) of the switches designated in the applicable paragraph headings. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.2.3.

4.4.2.3 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this section is identified in the specification sheet.

4.4.2.4 Sample size. A sample of eight (8) switches plus the quantity for life test (see 4.8) shall be selected from each lot and subjected to the tests listed in Table III.

4.5 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Room temperature. Room ambient of $23^{\circ}C \pm 5^{\circ}C$.

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b. Input power requirements. The switch shall perform satisfactorily under all applicable conditions specified herein from a 27.5 VDC rated voltage power source, with characteristics and limits as defined in MIL-STD-704.

4.6 Physical inspection.

4.6.1 Switch inspection. Each switch shall be thoroughly examined to determine conformance to all of the requirements of this specification and applicable documents for which no specific tests are described (see 3.28).

4.6.2 Packaging inspection. Each switch shall be examined to ensure that the packaging, packing and marking conforms to this specification (see Section 5).

4.6.3 Visual and mechanical examination. Switches shall be examined to verify that the materials, design, construction, physical dimensions, marking and workmanship are as specified herein.

4.7 Environmental tests. Prior to and after completion of each of the following tests, the switches shall be subjected to the electrical tests as specified in 4.9.1.

4.7.1 Humidity resistance. Switches shall be tested in accordance with Method 507, Procedure 1 of MIL-STD-810. The following details and exceptions shall apply:

a. Mounting. On a corrosion-resistant metal panel, by normal mounting means. Mating connector attached.

b. Initial measurement. The simulated firing test, 4.9.1, shall be performed.

c. At the end of cycle 5 and cycle 10, the sample shall be examined visually for damage or defects and the simulated firing test, 4.9.1, shall be performed.

d. Final measurements. Upon completion of step 6 of the final cycle, and while the switches are still in the humidity chamber, the simulated firing tests, 4.9.1, shall be performed. After a 24 hour drying period at a relative humidity of 50 $\pm$ 5 percent, switches shall be examined for evidence of corrosion, breaking, cracking, spalling and loosening of exterior components.

4.7.2 Salt fog (corrosion). Switches, with associated mounting hardware, shall be tested in accordance with Method 509 of MIL-STD-810, Procedure I. A 5 $\pm$ 1 percent salt solution shall be used. Immediately upon completion of the environmental conditioning, the sample shall be visually examined and the simulated firing test, 4.9.1, shall be performed. Following a 48 hour drying period, these tests shall be repeated. There shall be no evidence of excessive corrosion, peeling, chipping or blistering of the finish, nor exposure of base metal.

4.7.3 Sand and dust. The switch shall be tested in accordance with MIL-STD-810, Method 510. Immediately upon completion of the environmental conditioning, the

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sample shall be examined visually for damage or defects and the simulated firing test, 4.9.1, shall be performed.

4.7.4 Barometric pressure (reduced). The switch shall be tested in accordance with MIL-STD-202, Method 105, Test Condition B, 50,000 feet operating altitude for 4 hours minimum. Immediately upon completion of the environmental conditioning, prior to removal from the test chamber, the simulated firing test, 4.9.1, shall be performed.

4.7.5 Thermal shock. The switch shall be tested in accordance with MIL-STD-202, Method 107, Test Condition A-1 (11 cycles), except that the maximum temperature shall be +71°C. Operational test, 4.9.7, shall be conducted prior to the first cycle and upon completion of the final cycle. The first-five cycles shall be run continuously with a time of 1 hour for steps 1 and 3. During the next cycle, the switch shall be held at -55°C for 8 hours minimum and operational test, 4.9.7, conducted with temperature maintained. The test shall continue to and terminate at step 3 with the temperature remaining at +71°C for 8 hours minimum and operational test, 4.9.7, repeated with temperature maintained. The first-five cycles of the series shall be repeated for the conclusion of the series.

4.7.6 Shock test. The switch shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure IV, Figure 516.2-2, Amplitude A, Duration C. The switch shall be rigidly mounted to the shock machine table. For the environmental exposure, one switch shall be placed in the safe position with power applied, and the other switch shall be placed in the arm position with no power applied. Immediately upon completion of environmental conditioning, the samples shall be visually examined for damage or defects. Both switches shall be armed and the simulated firing test, 4.9.1, shall be performed. Any inadvertent advance of the switch or malfunction shall be cause for rejection. There shall be no evidence of broken, loose, deformed or displaced parts. Vibration isolators shall not be used.

4.7.7 Acceleration test. The switch shall be tested in accordance with MIL-STD-810, Method 513.2, Procedure II, Table 513.2-II. During each orientation in each axis, the simulated firing test, 4.9.1, shall be performed. Immediately upon completion of the environmental conditioning, the sample shall be visually examined for damage or defects and the simulated firing test, 4.9.1, shall be performed.

4.7.8 Vibration tests. The switch shall be tested in accordance with MIL-STD-810, Method 514.2, Procedure IIA. During a resonant dwell and at the end of vibration in each axis, a simulated firing test, 4.9.1, shall be performed on the switch. Immediately upon completion of the environmental conditioning, the sample shall be visually examined for damage or defects. There shall be no opening of grounded outputs or discontinuity of the firing output in excess of 10 microseconds (μs).

4.7.9 Electromagnetic compatibility tests. The switch shall be tested to the requirements of MIL-STD-461, as modified by 3.23 of this specification, and according to the measurement procedures defined by MIL-STD-462. The simulated firing test, 4.9.1, shall be used for operating the switch.

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4.7.10 Operating power test. The switch shall be tested to the transient voltage requirements of MIL-STD-704, as modified by 3.25 of this specification. The simulated firing test, 4.9.1, shall be performed, except that the transient voltage characteristics of MIL-STD-704, as modified by 3.25 of this specification, shall be superimposed on the input voltage.

4.8 Life test. Life test sampling quantity for production lots shall be as specified in Table IV. For first article inspection, one switch, each from Table I, sample Groups B and C, which have successfully passed all tests, shall be subjected to life testing. The contractor shall be responsible for conducting such tests to verify the operational life specified in 3.26. The tests shall be conducted on switches that have passed the individual inspections. The life test shall be performed under the conditions specified in 4.8.1.

4.8.1 Test conditions. The life test shall be conducted under the following conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5,000 feet)
Humidity	Room ambient

4.8.2 Test periods. Tests shall be conducted with the fire signal applied to the switch intermittently for an equal number of cycles of each life test, in accordance with Table I in the specification sheet. After completion of these tests, the tests of 4.9.3 and 4.9.4 shall be performed. The switch of this specification has one (1) operational cycle, per one (1) 360 degree shaft revolution. Means shall be provided to assure shaft rotation. Any failure shall be cause for rejection of the lot.

4.9 Electrical tests.

4.9.1 Simulated firing test. The switch shall be tested for proper timing, firing energy and operation with power supplied through a series combination of resistance and inductance. Upon application of power, the input voltage may be subjected to variations similar to switch contact bounce. A dummy load, consisting of resistance, shall be momentarily connected between each output pin and ground simulating firing the dispenser. Switch timing shall be measured for compliance with timing requirements of the specification sheet. The current through each output resistance shall be measured for compliance with output current, dwell time and firing sequence requirements. Any switch failing to comply with the requirements shall be cause for rejection of the lot. Tests as listed in Table I of the specification sheet shall be used for the simulated firing test.

4.9.2 Timing.

4.9.2.1 Dwell time. The dwell time shall be measured using an oscilloscope or other suitable method. The source voltage shall be the voltage specified in 3.21.5. The test circuit shown in the specification sheet shall be used. For purposes of test, the dwell time shall be exclusive of contact bounce. All firing outputs shall be observed.

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4.9.2.2 Externally-pulsed stepping speed. The ability of the switch to meet the specified pulse on and off time requirements, when externally pulsed, shall be checked by using an oscilloscope or other suitable method.

4.9.2.3 Initial delay time. The initial delay time shall be measured using an oscilloscope or other suitable method. The source voltage shall be the voltage specified in 3.21.5. The test circuit shown in the specification sheet shall be used. For purpose of test, the maximum initial delay time shall be exclusive of contact bounce; however, the minimum delay shall include contact bounce.

4.9.2.4 Contact bounce. A photographic record shall be made of an oscilloscope trace or other suitable means used to record contact bounce. Maximum contact bounce shall be as specified in 3.21.6.4.

4.9.2.5 Electrical circuit. The electrical test circuit in the specification sheet shall be utilized for all timing tests.

4.9.3 Safe circuit test. The stepping switch shall be tested for proper operation of the safe circuit using the following tests:

a. Set the switch to the safe position with the output pins unloaded. Prior to application of power, the resistance shall be measured between each of the firing output pins and the ground pin. The maximum resistance measured shall not exceed 0.2 OHMS.

b. Set the switch to the safe position. Dummy loads consisting of one ohm resistors shall be connected between each firing output pin and the ground pin. When 22 VDC (with 5 OHM series resistor) or 31.5 VDC (with 2.5 OHM series resistor) is applied, the switch shall not sequence out of the safe position. The voltage measured across each one OHM load shall not exceed 50 millivolts.

The switches shall be tested 100 percent for compliance with the safe circuit test requirements. Any switch failing to comply with the requirements shall be classified defective and shall be cause for rejection of the lot.

4.9.4 Arm circuit test. The stepping switch shall be tested for proper operation of the arm circuit using the following tests:

a. Set the switch to the arm position with the output pins unloaded. Prior to application of power, the resistance shall be measured between each of the firing output pins and the ground pin. The maximum resistance measured shall not exceed 0.2 OHMS.

b. Set the switch to the arm position with the output pins unloaded. Prior to application of power, manually advance the switch selector shaft through 360 degrees. The resistance shall be measured between each of the firing output pins and the ground pin. The maximum resistance measured shall not exceed 0.2 OHMS, except on the selected output as determined by the switch knob position on the indicator label.

c. Set the switch to the arm position. Dummy loads consisting of one OHM resistors shall be connected between each firing output pin and

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the ground pin. When 31.5 VDC (with 2.5 OHM series resistor) is applied, the switch shall sequence out of the arm position into firing position one. The voltage measured across each one OHM load, except at firing output number one, shall not exceed 50 millivolts.

The switches shall be tested 100 percent for compliance with the arm circuit test requirements. Any switch failing to comply with the requirements shall be classified defective and shall be cause for rejection of the lot.

4.9.5 Sequence test. The switch output firing order shall be checked during the sequence test in Table I of the specification sheet, to ensure that the output sequence is as specified in the specification sheet. Any switch failing to comply with these requirements shall be classified defective and shall be cause for rejection of the lot.

4.9.6 Bypass test. The switch shall be operated at 22 to 31.5 VDC with one or more output pin(s) shorted to ground and/or open circuited. The switch shall attempt to fire and remain in place with each application of power. The switch must not be damaged by a shorted output. The indicated tests in Table I of the specification sheet shall be used. Any switch failing to comply with the requirements shall be classified defective and shall be cause for rejection of the lot.

4.9.7 Operational test. The switch shall be tested for proper timing, firing energy and operation with power supplied at various voltages through various combinations of input line resistance and inductance, and with various dummy output loads. The input voltage shall be subjected to variations similar to contact bounce. Switch timing shall be measured for compliance with timing requirements. The current across each output resistance shall be measured for compliance with output current, dwell time and firing sequence requirements. The switches shall be tested 100 percent for compliance with all operational tests specified in Table I of the specification sheet. Any switch failing to comply with the requirements shall be classified defective and the lot shall be rejected. All tests of Table I of the specification sheet shall be used for operational test.

4.9.8 Steady state input current. With rated voltage applied to the switch and firing outputs unloaded, the current flow shall be determined within 30 seconds and shall not exceed the value specified in the specification sheet.

5. PACKAGING

5.1 Packaging, packing and marking. For shipment to a facility for immediate incorporation into a dispenser, the switch shall be packaged, packed and marked in accordance with Level C of MIL-STD-1169. For all other shipments, the packaging, packing and marking shall be Level A.

6. NOTES

6.1 Intended use. The switch is intended for use with removable aircraft store dispensers (as noted in the detail specification).

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet and part number.

b. Serialization requirements, if applicable.

c. Production lot acceptance, if other than specified in Section 4 of this specification.

d. Whether first article is required (see 3.2 and 6.3).

e. First article sample requirements, if other than specified in Section 4 of this specification.

f. Where first article samples are to be forwarded (see 4.3 and 6.3).

g. Samples subjected to lot inspection shall not be considered or accepted as part of the contract (see 4.4.2).

h. Lot inspection selection (see 4.4.2).

i. Name and location of Government quality assurance representative/ specialist responsible for random selection of inspection samples (see 4.4.2).

j. Name and location of Government approved test laboratory.

k. Packaging, packing and marking shall be as specified (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.
4.3, 4.4	Inspection and test reports	DI-T-5329

(Date item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DOD 5000.19L, Volume II, ADSDL. Copies of data item descriptions required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article is required for inspection and approval (see 3.2, 4.3, 6.2.1 and 6.3.1), the contract shall specify the following provision

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for first article inspection. When a contractor is in continuous production of switches from contract to contract consideration should be given to waive the first article inspections. If inspection is required, indicate:

a. If first article inspections are conducted at the contractor's plant or a Government approved laboratory, an inspection report shall be forwarded to the acquiring activity for verification.

b. That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3.1 First article information. Contracts shall specify the following provision for first article inspection. No production switches shall be delivered prior to the approval of the first article sample. Prefabrication of production switches prior to the approval of the first article sample is at the contractor's own risk. The approved first article sample shall be retained by the contractor for his own use in the fabrication and testing of switches to be submitted for acceptance. The first article sample shall not be considered as switches supplied under the contract for delivery as part of the production lot.

6.4 Quality assurance definitions. Definitions for quality assurance terms are in accordance with MIL-STD-109.

6.5 Definitions.

6.5.1 Contractor. Unless otherwise specified, the word "contractor" used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, subcontractors, Government-owned, contractor-operated (GOCO) and Government-owned, Government-operated (GOGO) plants.

6.5.2 Stepping switch. A stepping switch is defined as an electromagnetic device capable of switching one or more poles to any one of a predetermined number of positions. This being accomplished by the remote controlled rotation of the wiper assembly in a manner such that any pole can be made to contact one of its associated bank contacts. In addition, it may have interrupter contacts which operate and release once each rotary step and off-normal springs which operate at a prescribed point (or points) in the rotation.

6.5.3 Unidirectional, pulsed operation only. A unidirectional switch for pulsed operation only, is one designed to rotate in one direction only and required to take one step for each application of operating power.

6.5.4 Direct drive. A directly-driven stepping switch is one designed to advance its wipers when the circuit is closed to the actuating motor magnet.

6.5.5 Rated input voltage. The design voltage which may be applied for the rated duty cycle.

6.5.6 Cycle. A cycle is that number of steps required for the wiper to make

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and break each contact on the bank, i.e., 10 steps for a switch having a 10 point bank: 26 steps for a switch having a 26 point bank.

6.5.7 Switching contacts. Switching contacts are the contacts that are located on all levels and bank terminal positions that make or break due to wiper arm action or movement.

6.5.8 Auto-step. Auto-step is the condition in which the switch advances to the next firing station even though the fire input voltage remains valid.

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TABLE I. First article inspection.

Examination or test 1/	Paragraph	
	Requirement	Method
SAMPLE GROUP A (all units):		
Visual and mechanical examination (external and internal) 2/	3.4 thru 3.18, 3.27 and 3.28	4.5a and 4.6
Simulated firing	3.24	4.9.1
Timing	3.21.6	4.9.2
Operational characteristics	3.20	4.9.7
Safe circuit	3.20.1, 3.20.5 and 3.21.3	4.9.3
Arm circuit	3.20.2, 3.20.5 and 3.21.3	4.9.4
Sequence	3.20.3	4.9.5
Bypass capabilities	3.20.4	4.9.6
Steady state input current	-	4.9.8
Unused pins	3.21.4	-
SAMPLE GROUP B (3 units):		
Humidity	3.22a	4.7.1
Salt fog (corrosion)	3.22b	4.7.2
Sand and dust	3.22c	4.7.3
Barometric pressure (reduced)	3.22d	4.7.4
SAMPLE GROUP C (3 units):		
Thermal shock	3.22e	4.7.5
SAMPLE GROUP D (3 units):		
Shock	3.22f	4.7.6
Acceleration	3.22g	4.7.7
Vibration	3.22h	4.7.8
SAMPLE GROUP E (2 units):		
Electromagnetic compatibility	3.23	4.7.9
Operating power	3.25	4.7.10
LIFE TESTS (2 units): (One switch from Group B and one switch from Group C)	3.26	4.8

1/ Units shall successively, and in the order listed, undergo the tests required for the sample groups to which they have been assigned.

2/ When sealed switches are submitted, the internal examination shall be conducted on an unsealed unit. This sample unit shall not be used for any subsequent test.

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TABLE II. Individual inspections.

Examination or test	Paragraph	
	Requirement	Method
SAMPLE GROUP (all units):		
Physical inspection	3.4, 3.6, 3.12, 3.13, 3.15 thru 3.18, 3.27 and 3.28	4.4.1.1 and 4.5a
Safe circuit	3.20.1, 3.20.5 and 3.21.3	4.9.3
Arm circuit	3.20.2, 3.20.5 and 3.21.3	4.9.4
Manufacturing run-in test	3.20	4.4.1.3 and 4.9.7

TABLE III. Lot inspections.

Examination or test 1/	Paragraph	
	Requirement	Method
SAMPLE GROUP A (all units):		
Visual and mechanical examination (external and internal) 2/	3.4 thru 3.18, 3.27 and 3.28	4.5a and 4.6
Simulated firing	3.24	4.9.1
Timing	3.21.6	4.9.2
Operational characteristics	3.20	4.9.7
Safe circuit	3.20.1, 3.20.5 and 3.21.3	4.9.3
Arm circuit	3.20.2, 3.20.5 and 3.21.3	4.9.4
Sequence	3.20.3	4.9.5
Bypass capabilities	3.20.4	4.9.6
Steady state input current	-	4.9.8
Unused pins	3.21.4	-
SAMPLE GROUP B (2 units):		
Humidity	3.22a	4.7.1
Salt fog (corrosion)	3.22b	4.7.2
Barometric pressure (reduced)	3.22d	4.7.4
SAMPLE GROUP C (2 units):		
Thermal shock	3.22e	4.7.5
SAMPLE GROUP D (2 units):		
Shock	3.22f	4.7.6
Vibration	3.22h	4.7.8
LIFE TESTS 3/	3.26	4.8

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TABLE III. Lot inspections. - Continued

1/ Units shall successively and in the order listed undergo the tests required for the sample groups to which they have been assigned.

2/ When sealed switches are submitted, the internal examination shall be conducted on an unsealed unit. This sample unit shall not be used for any subsequent test.

3/ Quantity of production switches to be tested as per 4.8 of this specification.

TABLE IV. Life test sampling.

Quantity of switches offered for acceptance per contract	Quantity to be selected for life test
First 25	1
Next 100	1
Next 250	1
	1 for each additional 500 or fraction thereof

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